

Solmaz Haddady, MSc

Machine Learning Engineer | Computer Vision | Deep Learning |
Multimodal AI Scientific & Medical Data

Linz, Austria

+43 6606408565

solmazhaddadi4003@gmail.com

LinkedIn: [linkedin.com/in/solmaz-haddady-8a94801a3](https://www.linkedin.com/in/solmaz-haddady-8a94801a3)

GitHub: <https://github.com/solmazhaddady>



PROFESSIONAL PROFILE

AI Researcher and Machine Learning Engineer with a strong quantitative and interdisciplinary background spanning Artificial Intelligence, Bioinformatics, Geophysics and Physics. Hands-on experience developing end-to-end AI and data pipelines for complex real-world datasets, from data preprocessing, quality control and feature extraction to model development, evaluation and multimodal learning.

Experience spans computer vision, deep learning, Vision–Language Models, large-scale image and time-series data, signal processing, anomaly detection, scientific modelling and data standardization. Experienced working with interdisciplinary domain experts and translating complex data and research problems into reproducible computational solutions. Interested in AI/ML, computer vision, scientific ML and data-driven engineering across research and industry.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer & Medical AI Researcher

Department of Dermatology and Venereology, JKU Linz | 2023 – 2025

- Built preprocessing, quality-control, and feature-extraction workflows for approximately 3,000 whole-slide images, including heterogeneous clinical metadata and dataset standardization.
- Developed end-to-end deep learning pipelines for large-scale medical image analysis using real-world dermatopathology data.
- Designed CNN–Transformer models for whole-slide image classification and rare-pattern recognition, achieving 96.65% validation accuracy.
- Applied model evaluation, statistical validation, and benchmarking to assess performance, robustness, and reproducibility.
- Developed a multimodal Vision–Language Model pipeline integrating WSI representations with large language models for automated histopathology report generation.
- Integrated heterogeneous medical imaging and structured clinical metadata into unified machine-learning workflows.
- Collaborated with dermatologists, pathologists, and clinical researchers to translate clinical requirements into computational AI solutions.
- **Research Contributions**
 - Contribution to preparation and curation of a dermatopathology dataset for public release on The Cancer Imaging Archive (TCIA)

AI Research Assistant – Digital Pathology

Department of Pathology & Molecular Pathology, JKU Linz | 03/2023 –10/2023

- Contributed to AI methods for **computer-assisted diagnosis and tumor regression grading** in digital pathology.
- Processed, organized and validated clinical imaging datasets for machine-learning workflows.
- Supported data annotation, validation and preparation under real-world clinical constraints.
- Collaborated within an interdisciplinary research team involving AI and medical experts.

Seismic Data Processing & Analysis

OEOC Co., Tehran, Iran | 2014 – 2018

- Processed and analyzed **large-scale seismic time-series data** for gas-detection and geological interpretation applications.
- Applied **signal processing and time-frequency analysis** to identify relevant signal patterns and characteristics.
- Performed data preprocessing, noise reduction, seismic interpretation and quantitative pattern analysis.
- Investigated anomalous signal patterns and applied **simulation and numerical modelling** to analyze gas-related seismic responses.
- Used R and domain-specific scientific software for data processing, analysis and interpretation.

Data Management & Quality Control

- Supported **quality control, validation and standardization** of large scientific metadata collections.
- Performed consistency checks and metadata standardization to improve data quality and downstream usability.
- Supported **noise-detection and QC workflows** for large-scale scientific datasets.
- Worked with data-management processes involving organization, validation and preparation of heterogeneous datasets.

Tutor for Physics & Mathematics

Self-Employed, Tehran, Iran | 2014 – 2017

- Instructed university and high school students in mathematics and physics.

SELECTED AI PROJECTS

DermPath-LLaMA

Multimodal Vision–Language Model for Histopathology Report Generation

Python | PyTorch | Transformers | LoRA | LLaMA

- Developed an end-to-end multimodal AI system integrating computer vision and large language models.
- **Repository:** https://github.com/solmazhaddady/DermPath_LLaMA_VLM

NMSC Whole-Slide Imaging Dataset

Large-Scale Medical Imaging_Text Dataset & TCIA Preparation

Python | WSI | DICOM | OpenSlide | Pandas

- Curated and standardized approximately 3,000 whole-slide images and associated metadata for AI research and public dataset preparation.
- Built reproducible **preprocessing, quality-control and metadata-validation workflows** for large-scale medical imaging data.
- **Repository:** <https://github.com/solmazhaddady/NMSC-TCIA-Dataset>

WSI-to-DICOM Converter

Medical Imaging Interoperability Pipeline

Python | DICOM | OpenSlide | wsidicomizer

- Developed an automated pipeline for converting proprietary whole-slide image formats (SVS, MRXS) into the DICOM standard.
- Improved interoperability between digital pathology platforms and AI research workflows.
- Designed reusable software components supporting standardized medical imaging research.
- **Repository:** github.com/solmazhaddady/WSI-to-DICOM-Converter

Whole-Slide Image Quality Control Pipeline

Automated Image Quality Assessment Pipeline

Python | Computer Vision | Image Processing

- Developed automated preprocessing and quality-control workflows for large-scale datasets.
- Implemented image validation, metadata verification, and data curation utilities supporting reliable AI training datasets.
- Designed reusable tools for scalable data preparation.

EDUCATION

Master of Science in Artificial Intelligence

Johannes Kepler University (JKU), Linz, Austria | 2022 – 2025

- Specialization: Machine Learning, Deep Learning, Computer Vision, Medical Imaging
- Thesis: Classification and Vision–Language Modeling for Skin Cancer Histopathology (Grade: excellent)
- Presented research at the **9th & 10th ÖGDV Science Days (2024 & 2025), Austria**

Bachelor of Science in Bioinformatics

Johannes Kepler University (JKU), Linz, Austria | 2017 – 2022

- Interdisciplinary studies combining Computer Science and Molecular Biology
- Erasmus+ Exchange: University of South Bohemia, Czech Republic.

Master of Science in Geophysics:

Tehran University, Iran | 2015 – 2017

- Focus on numerical modeling, wave physics, and earth sciences.
- Thesis: Gas and low-frequency shadows detection with S-transformer time-frequency method (Grade: excellent)

Bachelor of Science in Physics

Tehran, Iran | 2012 – 2015

- Foundational studies in theoretical physics and mathematics.

RESEARCH EXPERIENCE

- **International Summer School on Seismology, Shiraz University, Iran | 2016**
- **Poster Presentation – 9th ÖGDV Science Days, Bad Gastein, Austria (2024)**
- **Poster Presentation – 10th ÖGDV Science Days, Werfenweng, Austria (2025)**

TECHNICAL SKILLS

Programming

- Python • R

AI & Machine Learning

- Python • PyTorch • TensorFlow • scikit-learn • Deep Learning • CNNs • Transformers • Vision Transformers • VLMs • LLMs • Multimodal Learning • Generative AI

Computer Vision

- Computer Vision • Image Processing • Feature Extraction • Classification • Medical Image Analysis • Whole-Slide Imaging

Medical Imaging

- Whole-Slide Imaging (WSI) • DICOM • Digital Pathology • Medical Image Analysis • OpenSlide • Wsidicomizer

Data Science & Scientific Computing

- Pandas • NumPy • R • Data Preprocessing • Data Curation • Quality Control • Statistical Validation • Signal Processing • Time-Series Analysis • Time-Frequency Analysis • Anomaly Detection • Numerical Modelling

Data & Interoperability

- Metadata Validation • Data Standardization • Data Integration • Large-Scale Data Pipelines

Engineering & Research

- Git • Linux • Docker • CUDA/GPU Computing • Jupyter • VS Code • PyCharm • Model Evaluation • Benchmarking • Reproducible Research

SELECTED PUBLICATION & MANUSCRIPTS

Haddady, S., et al. (Manuscript in preparation).

“A High-Resolution Dataset of Non-Melanoma Skin Cancer Whole-Slide Images for Machine Learning Research” — planned submission to *Scientific Data* (Nature Portfolio).

LANGUAGES

- English: Professional Working Proficiency
- German: +B2
- Persian: Native